

★ 16-WEEK SYLLABUS

Financial Machine Learning

Undergraduate 3rd–4th Year · Beginner-Friendly Curriculum

Wed 18:00–20:00 Online (recorded) · Fri 10:00–11:00 In-Person Q&A

Built on López de Prado, Hull, and Causal Factor Investing — covering product recommendation, fraud detection, and AI chatbots from traditional models to the frontier.

Course Overview

Course information at a glance

Item	Details
Course	Financial Machine Learning
Target	Undergraduate 3rd–4th year (beginners welcome)
Duration	16 units over 14 teaching weeks · 3 hrs/week
Language	Python — scikit-learn, XGBoost, PyTorch
Format	Primer + Wed online (recorded) + Fri in-person Q&A

Three applications — Product Recommendation · Fraud Detection · AI Chatbot

Weekly Schedule — Online Lecture + In-Person Q&A

Watch the unit primer first · Wednesday online · Friday in the classroom

Day	Time	Format	What happens
Wednesday	18:00–20:00	Online · recorded	Full lecture + code walkthrough
Friday	10:00–11:00	In-person · classroom	Q&A, lab review, project feedback

Before Friday

- Watch the recording in full
- Re-run the notebook yourself
- Write down what broke

In the Friday hour

- Your questions set the agenda
- No re-teaching of the lecture
- We debug and extend live

Assessment & Examinations

Continuous evaluation, two written examinations

30%

Weekly
assignments

20%

Midterm exam
(Week 8)

30%

Final exam
(Week 16)

20%

Participation
(Friday hour)

Midterm — week 8, units 1–8 · Final — week 16, units 9–16

Participation is earned in the Friday hour — attendance plus the questions you bring

Textbooks & References

2 primary · 3 supplementary

Primary textbooks

- [Hull] Hull et al. (2025), Machine Learning in Business, 4th ed.
- [AFML] López de Prado (2018), Advances in Financial Machine Learning

Supplementary

- [MLAM] Machine Learning for Asset Managers (2020)
- [CFI] Causal Factor Investing (2023)
- [CFIP] Causality and Factor Investing: A Primer (2025)

Chosen for what makes financial data hard — non-stationarity, low signal-to-noise, overlapping labels

4-Part Curriculum Structure

Foundations → Core Algorithms → Financial ML in Depth → Frontier

Part	Units	Focus
I · Foundations	1–3	ML paradigms · Bars & CUSUM · Triple-Barrier · Meta-Labeling
II · Core Algorithms	4–9	Linear · Logistic/KNN/NB · SVM · Trees & Ensembles · DL · Unsupervised
III · ML in Depth	10–13	Purged CV · DSR/PBO · HRP/NCO · NLP/LLM/RAG · Reinforcement Learning
IV · Frontier	14–16	Causal inference · SHAP/XAI · Fairness · TFT/GNN/AI Agent

Midterm exam in week 8 (units 1–8) · Final exam in week 16 (units 9–16)

Sixteen Units, Fourteen Teaching Weeks

Weeks 8 and 16 are the examinations — so two weeks carry two units

Week	Unit · Topic
1	U1 · ML and the Finance Problem
2	U2 · Financial Data Structure
3	U3 · Labeling and Meta-Labeling
4	U4 · Linear Models and Regularization
5	U5 + U6 · Classifiers and SVM
6	U7 · Trees and Ensembles
7	U8 · Neural Networks
8	MIDTERM — units 1–8

Week	Unit · Topic
9	U9 · Unsupervised Learning
10	U10 · Honest Backtesting
11	U11 · Portfolio Construction
12	U12 · NLP, LLMs and RAG
13	U13 · Reinforcement Learning
14	U14 + U15 · Causality and Frontier
15	U16 · Integration and Capstone
16	FINAL — units 9–16

Week 5 pairs the three simple classifiers with SVM; week 14 pairs causality with the frontier. Units 1–8 are examined in week 8, units 9–16 in week 16.

Part I — Foundations

ML basics → Financial data → Labeling

Unit	Topic	Key content	Texts
1	ML & Finance	Bias–variance tradeoff · Sisyphus vs. meta-strategy Traditional models: CF/MF, rule-based fraud & chatbot	[Hull] Ch.1 [AFML] Ch.1
2	Financial Data	Time / Volume / Dollar Bars · CUSUM filter Data cleaning · outliers · survivorship bias	[Hull] Ch.1 [AFML] Ch.2
3	Labeling	Triple-Barrier method · Meta-Labeling Precision / recall control · class imbalance	[AFML] Ch.3–4

Deep learning recommenders (NCF, Two-Tower, Sequential, GNN) are previewed in unit 1

Episodes & Case Studies

Why each week's idea mattered outside the classroom

1

U1 · LTCM vs. Renaissance

The same mathematics, opposite fates — Sisyphus versus meta-strategy.

2

U2 · The 2010 Flash Crash

Garbage in, garbage out — data quality decides whether a model survives.

3

U3 · The wrong question

"Will the price go up tomorrow?" — event-based labeling asks a better one.

Every part opens with the story before it opens with the algorithm

Part II-A — Core Algorithms

Linear models → three classifiers → SVM

Unit	Topic	Key content	Anomaly	Texts
4	Linear & Regularization	OLS · Ridge · Lasso · Elastic Net Cross-validation · index tracking	—	[Hull] Ch.3 [MLAM] Ch.2–3
5	Logistic, KNN, NB	Logistic (PD) · KNN · Naive Bayes Discriminative / instance / generative	LOF	[Hull] § 3.9, § 3.12
6	SVM	Hard & soft margin · RBF kernel SVR for volatility	OCSVM	[Hull] § 5.1–5.4

Episodes — U4 the Factor Zoo · U5 birth of the FICO score · U6 Vapnik's journey

Where Units 4–6 Show Up

The same three applications, one layer deeper each week

Product Recommendation

U4–6

KNN finds similar customers
Logistic gives purchase probability

Fraud Detection

U4–6

NB first filter → Logistic second
LOF local density → OCSVM boundary

AI Chatbot

U4–6

TF-IDF + SVM intent classification
then the transition to BERT

Traditional baselines are taught first, then replaced only when the data justifies it

Part II-B — Core Algorithms

Trees & ensembles → deep learning → unsupervised

Unit	Topic	Key content	Anomaly	Texts
7	Trees & Ensembles	Decision Tree · Random Forest · XGBoost Feature importance: MDI / MDA / SFI	Isolation Forest	[Hull] Ch.4 [AFML] Ch.6–8
8	Neural Nets & DL	FF-NN · CNN (vol surface) · LSTM NCF · Two-Tower · Sequential Rec	Auto-encoder	[Hull] Ch.6 [AFML] Ch.18
9	Unsupervised	K-Means · PCA · Random Matrix Theory Marchenko–Pastur · denoised correlation	PCA anomaly	[Hull] Ch.2 [MLAM] Ch.2–4

Episodes — U7 why XGBoost keeps winning · U8 the J.P. Morgan confession · U9 physics to finance

Where Units 7–9 Show Up

Ensembles, deep learning, and unsupervised structure in production

Product Recommendation

U7–9

LightGBM purchase probability
NCF / Two-Tower ·
segmentation

Fraud Detection

U7–9

XGBoost + Isolation Forest
cascade
Autoencoder · LSTM sequence

AI Chatbot

U7–9

LSTM holds multi-turn context
across a full conversation

The midterm exam (week 8) draws directly on these three tracks

Part III — Financial ML in Depth

Cross-validation · portfolio · NLP/LLM · reinforcement learning

Unit	Topic	Key content	Texts
10	Financial CV & Backtesting	Purged k-fold CV · CPCV · Deflated Sharpe Ratio Probability of Backtest Overfitting (PBO)	[AFML] Ch.7, 9, 11–12
11	Portfolio Optimization	Limits of MVO · HRP · NCO Denoised covariance · robo-advisor	[MLAM] Ch.4–7 [AFML] Ch.16
12	NLP · LLM · Chatbot	TF-IDF → Transformer · FinBERT · RAG Chatbot evolution: rules → NLU → LLM+RAG	[Hull] Ch.8
13	RL & Optimal Execution	MDP · Q-Learning · DQN · LOXM RL sequential rec · dialogue strategy	[Hull] Ch.7 [AFML] Ch.18

Midterm Examination — Units 1–8

Ninety minutes, forty questions, every number taken from the lectures

① 40 Questions Multiple Choice

Format

Four options, exactly one correct.
No penalty for a wrong answer.

② Foundations → Neural Nets

Coverage

Data, labels, linear models, classification, trees, neural nets.

③ Hand Calculations

Preparation

Every number is taken from a lecture worked example.

Sat in week 8 · $40 \times 2.5 = 100$ points · 20% of the final grade

Part IV — Frontier & Causality

Causal inference · interpretation · latest trends · integration

Unit	Topic	Key content	Texts
14	Causality · XAI · Ethics	DAG · SCM · factor mirage · do-calculus SHAP · LIME · PDP · EU AI Act	[CFI] · [CFIP] [Hull] Ch.9–10
15	Trends & Integration	TFT · Diffusion · GNN (LightGCN, ContextGNN) AI Agent (LLM + tool use) · pipeline review	Papers [AFML] Ch.20
16	Integration & Capstone	One honest, end-to-end pipeline, reviewed Revision before the final exam	—
Episodes — U14 the Apple Card bias case · U15 augmentation, not replacement			

Final Examination — Units 9–16

The same format again, over the second half of the course

Coverage

- ① U9–10 honest testing
- ② U11–12 portfolios, LLMs
- ③ U13–14 RL and causality

Also examined

- ④ U15–16 the frontier
- ⑤ The whole pipeline
- ⑥ Where stages fail

Format

- ⑦ 40 questions · 90 min
- ⑧ 2.5 points each
- ⑨ No wrong-answer penalty

Sat in week 16 · 30% of the final grade

Three Industry Applications — Traditional to Frontier

One evolution path per application, deepened every week

Product Recommendation

- Rules → CF / MF
- KNN · Logistic · LightGBM
- NCF · Two-Tower · Sequential
- Robo-advisor (HRP)
- LLM · RL · GNN

Fraud Detection

- Rules + Z-score
- Logistic scorecard
- NB · LOF · OCSVM · XGBoost
- Isolation Forest · Autoencoder
- SHAP · Diffusion · TFT

AI Chatbot

- Rule-based dialogue tree
- TF-IDF + SVM intent
- BERT NLU · LSTM context
- LLM + RAG
- RL strategy · AI Agent

Anomaly Detection — Four Methods Compared

LOF → OCSVM → Isolation Forest → Autoencoder, each catching more complex patterns

Method	Unit	Approach	Strength	Limitation
LOF	U5 · KNN	Local density	Handles uneven density Customer-context anomaly	$O(n^2)$, small data only
OCSVM	U6 · SVM	Decision boundary	Stable in high dimension No labels needed	$O(n^2-n^3)$ cost
Isolation Forest	U7 · Tree	Path length	$O(n)$ fast Large-scale ready	Weak on local anomalies
Autoencoder	U8 · DL	Reconstruction	Nonlinear patterns VAE for augmentation	High training cost

Production Cascading Architecture

Screen wide, score precisely, then explain

1st · Screening

Unsupervised

Isolation Forest / OCSVM
No labels, catches novel fraud

2nd · Scoring

Supervised

XGBoost + LSTM
Sequence patterns, high
precision

3rd · Explaining

Compliance

SHAP + manual review
EU AI Act, regulatory audit

No single detector is enough — the architecture is the answer

Prerequisites & Learning Environment

Self-study resources for beginners — start before Week 1

Area	Minimum level	Recommended resource
Python	pandas, numpy, matplotlib basics	Kaggle Learn Python
Probability	Conditional probability, Bayes, testing	Khan Academy Statistics
Linear Algebra	Matrix multiplication, eigendecomposition	3Blue1Brown — Essence of Linear Algebra
Finance	Returns, risk, portfolio theory basics	Bodie et al., Investments Ch.1–8

None of these are graded — they are what makes Week 1 comfortable

Four Special Sessions — Beyond the Sixteen Units

Optional and self-contained — click a session, or browse keerhee.github.io/financial-machine-learning

Session	Slides	Theme	What it covers	Builds on
S1 · Recsys	23	Classic	Content-based and collaborative filtering	U9 factors · U12 cosine
S2 · Recsys 2	30	Deep	Factorization machines, then graphs and LLMs	U8 nets · U15 GNN
S3 · AI Firm	26	Agents	The firm as a team of cooperating agents	U10–U15 together
S4 · LangGraph	28	Build	That firm written as an explicit state machine	U13 RL · U12 LLM

Financial ML Is a Pipeline

Every week adds one stage to the same pipeline

**"Financial ML is not a single model — it is a pipeline."
Marcos López de Prado**

End-to-end pipeline

- Data → Bars → Labeling → Features → Purged CV → Training → SHAP → DSR
backtest → RL execution

Wed 18:00–20:00 online (recorded) · Fri 10:00–11:00 in person — bring your questions